ALCHEMUS

Advanced Drug Analysis Platform

Research Owner Wallet

A8ov5UqDdHWUAr8J81BCdUdoewNAPaGAtyH7TvQTYPoM

Date: 5/1/2025
Compounds: Amoxicillin, Dextromethorphan

New Compound Analysis

Name: Amoxidextro
Formula: $C_{41}H_{52}N_4O_{10}S$
Weight: 808.95
Drug-likeness: 0.72

Activity:
Broad-spectrum antimicrobial and antitussive

Target Analysis

Penicillin-binding proteins (PBPs) in bacteria
Binding Affinity: -9.5
Essential for bacterial cell wall synthesis

Sigma-1 receptor in the CNS
Binding Affinity: -8.2
Modulates cough reflex and CNS symptoms

Toxicity Profile

Risk Level: Moderate
Affected Systems: Liver, Central nervous system, Skin

Mitigation Suggestions:

- Dose monitoring
- Liver function tests
- Patient history review for allergies

Pharmacokinetics

Absorption:
Rapid oral absorption

Distribution:
Widely distributed, crossing the blood-brain barrier

Metabolism:
Hepatic via cytochrome P450 enzymes

Excretion:
Renal and fecal

Half-life: 4 to 6 hours

Detailed Analysis

Amoxidextro is a synthesized compound composed of both amoxicillin and dextromethorphan moieties, designed to exploit their complementary actions in a unified molecular structure. This new compound capitalizes on the beta-lactam structure of amoxicillin to exert antibacterial effects, chiefly inhibiting penicillin-binding proteins essential to bacterial cell wall biosynthesis. Meanwhile, the dextromethorphan part acts primarily on sigma-1 receptors in the central nervous system, providing antitussive effects. Such a combination is foreseen to be particularly beneficial in treating conditions where cough accompanies bacterial infections, like certain types of bronchitis or pneumonia. However, close attention must be paid to its toxicity profile, given the involvement of hepatic metabolism and the potential for allergic reactions. Preclinical studies will focus on mapping out any Section pharmacokinetic peculiarities and identifying any untoward syergistic toxicities.